



Expernetic (Pvt) Ltd

Talent Assessment Program · Technical Assignment

Data Engineer Intern

Assignment

Summary: Completed Work

This assignment was approached with a focus on establishing a high-quality, production-ready foundation for the Airbnb market intelligence analysis. The following sections were completed:

- **Dataset Familiarization:** A comprehensive exploration of the Inside Airbnb dataset was conducted to establish a baseline. This included schema documentation, mapping relationships between files (listings, calendar, reviews), and documenting data limitations and domain-specific assumptions. This mandatory step served as the technical foundation for all subsequent tasks.
- **Data Engineering Challenge:** A repeatable, clean, and scalable ingestion and cleaning pipeline was implemented. This involved data profiling, validation against domain-specific constraints (e.g., price and occupancy logic), and the design of a dimensional model suitable for analytical querying.
- **Exploratory Data Analysis (EDA):** A multi-faceted analysis was performed, covering statistical distributions, geographic pricing gradients, and temporal trends. Every visualization and statistical finding was paired with a business interpretation, transforming raw data into actionable market insights.

Summary: Incomplete Work

In alignment with the assignment's design philosophy, which emphasizes that "quality outweighs quantity," I made a strategic decision to scope my effort to core engineering and analytical tasks. The following sections were prioritized over optional modules:

- **Rationale for Prioritization:** The assessment duration occurred simultaneously with active academic obligations. To manage this constraint effectively, I prioritized deep execution of foundational tasks over the breadth of optional advanced modules (such as Cloud-Native deployment, Advanced ML, or AI/LLM experimentation).
- **Decision-Making:** I chose to treat the "Dataset Familiarization," "Data Engineering," and "EDA" sections as the critical path. I reasoned that demonstrating proficiency in raw data handling, pipeline robustness, and clear business storytelling is more indicative of long-term success in a Data Engineering role than attempting advanced, optional topics superficially.
- **Scope Boundaries:** By limiting the scope, I was able to dedicate sufficient time to the "Engineering Decision Log," ensuring that my approach, trade-offs, and methodology were documented with the depth expected of senior technical staff, rather than rushing through elective sections that would have diluted the quality of the primary submission.